

Product Recommendation System

Ralph Djino DUMERA & Rockson NOEL

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92.98%

F1 Score

SVM

Model Performance

SVM	92.98%
Random Forest	90.52%
Gradient Boosting	88.49%
Logistic Regression	87.51%
Naive Bayes	85.47%



OVERVIEW

Goal:

Develop a hybrid recommendation system that leverages customer sentiment from textual reviews to enhance product relevance.

Objectives:

- Analyze customer reviews using NLP.
- Classify sentiment polarity (positive/negative).
- Integrate sentiment predictions into recommendation logic.

Key Metric: F1-Score.



Business Problem

Challenge:

Traditional recommender systems rely mainly on numeric ratings and overlook emotional tone. This can lead to *irrelevant or misleading product suggestions*.

Solution:

Integrate **sentiment analysis** to capture the emotional context and improve recommendation quality.

“Ratings tell how much users like a product. Sentiment tells *why*.”



Dataset Overview

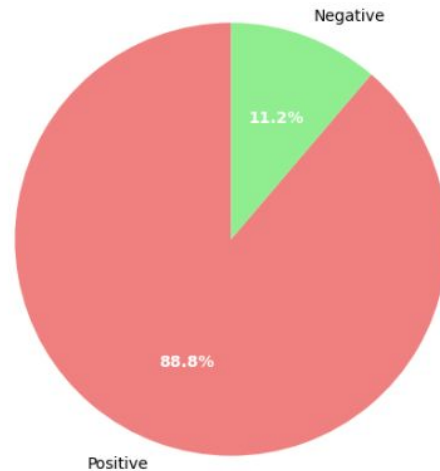
Source: Kaggle – *Sentiment-Based Product Recommendation System*

Size: ~30,000 product reviews

Key Columns:

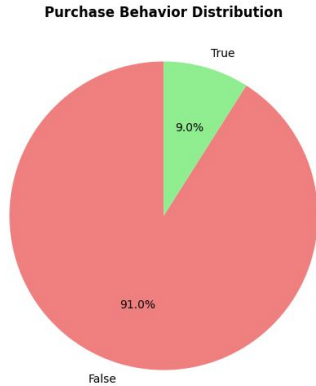
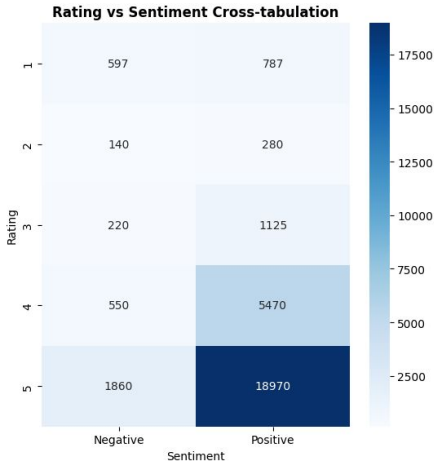
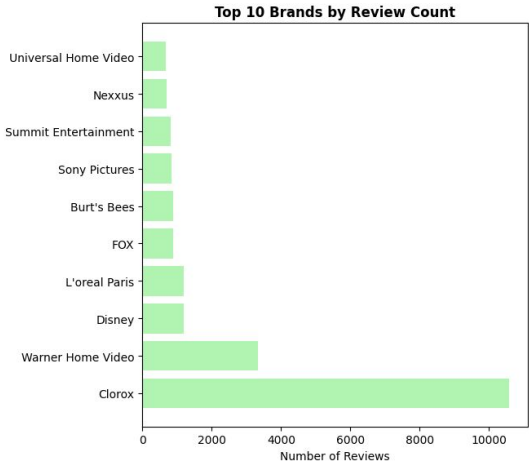
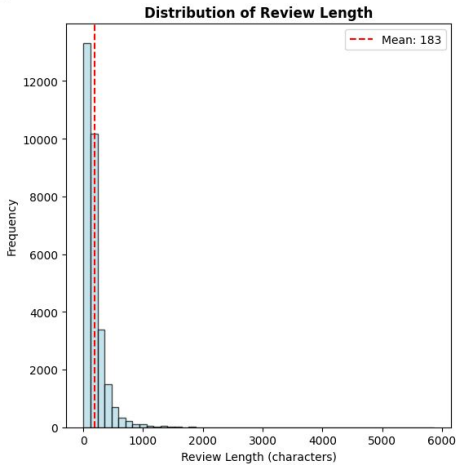
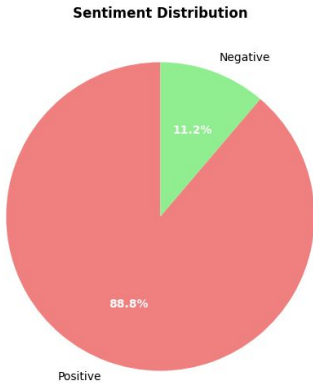
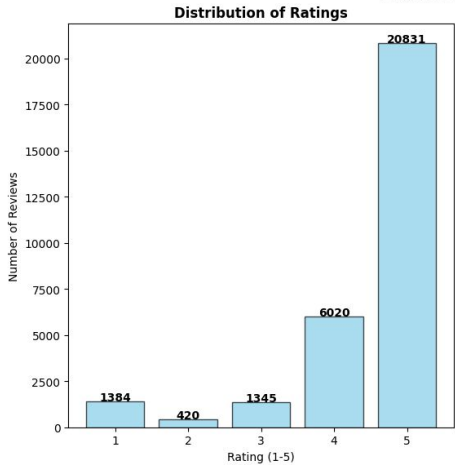
`name`, `brand`, `categories`, `reviews_text`, `reviews_rating`,
`user_sentiment`

Sentiment Distribution



EDA (Exploratory Data Analysis)

Dataset Overview - Basic Statistics and Distributions



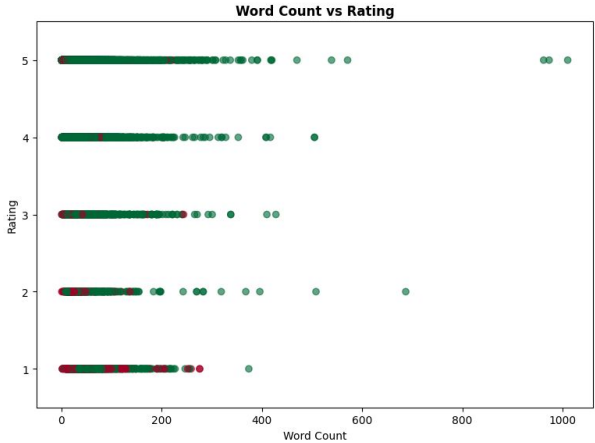
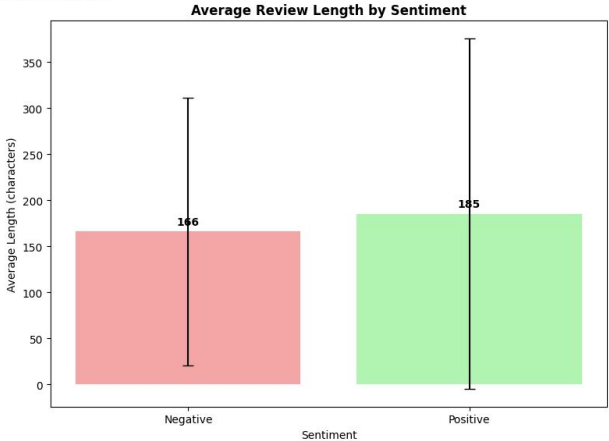
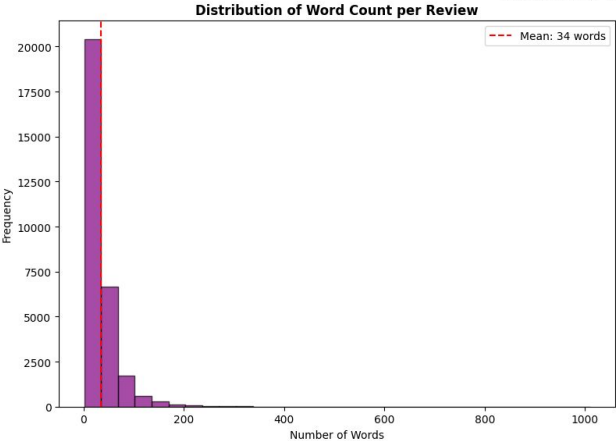


EDA (Exploratory Data Analysis)

- Most reviews have **4★ or 5★ ratings**.
- Texts are short (avg. 183 characters).
- A few brands dominate (Clorox, Warner Home Video).
- Positive sentiment prevails across time.

Advanced Text Analysis

Advanced Text Analysis





Advanced Text Analysis

- Reviews average **34 words**, confirming short-form opinions.
- Negative reviews are slightly longer and more detailed.
- High variance in low ratings (1–2★) → emotionally charged feedback.



Positive Reviews Word Cloud





World Cloud Insights

Positive Sentiment:

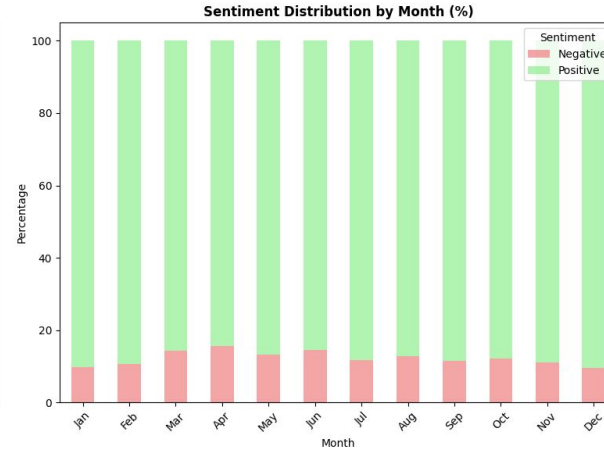
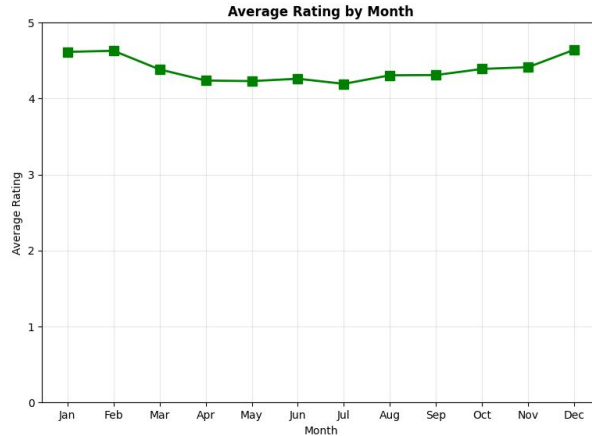
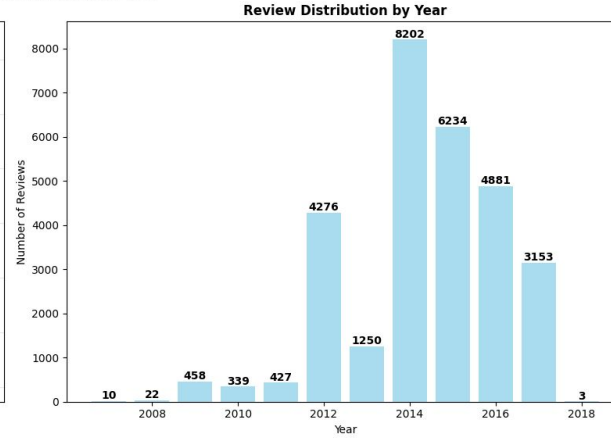
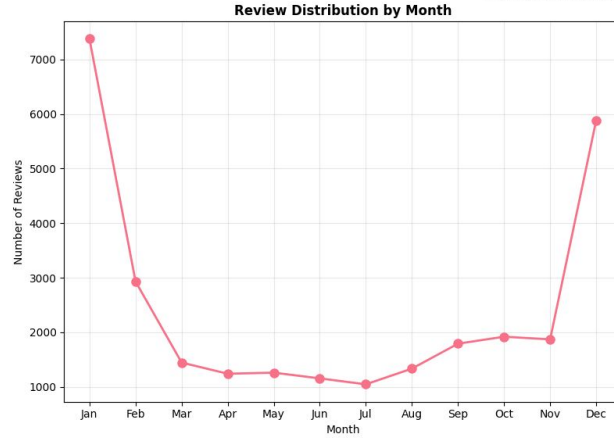
Words like *love, great, clean, easy, use* → satisfaction and usability.

Negative Sentiment:

Words like *disappointed, smell, old, changed* → quality issues and dissatisfaction.

Temporal Analysis

Temporal Analysis of Reviews



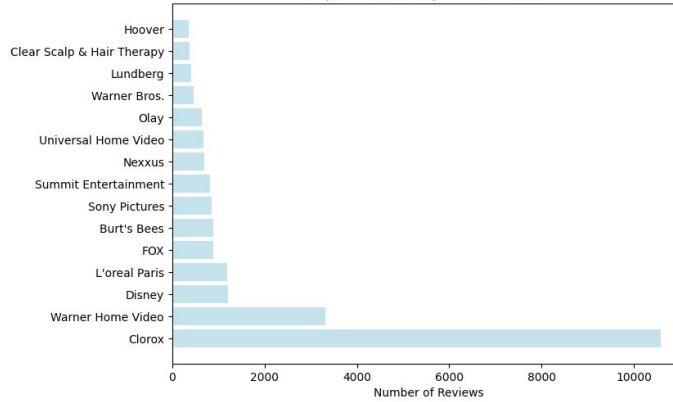


- Review peaks in **January and December** (holiday effect).
- Ratings stable around **4.3–4.7** across the year.
- Positive sentiment dominates every month (>85%).

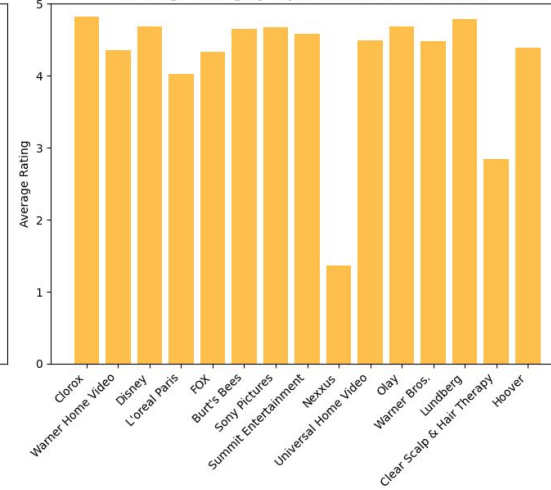
Brand and Category Insights

Brand and Category Analysis

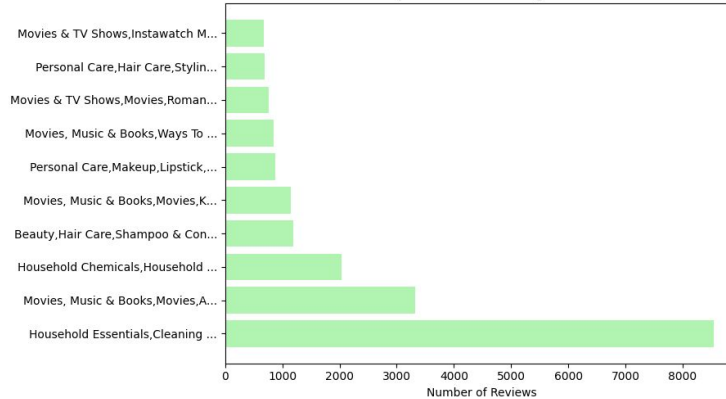
Top 15 Brands by Review Count



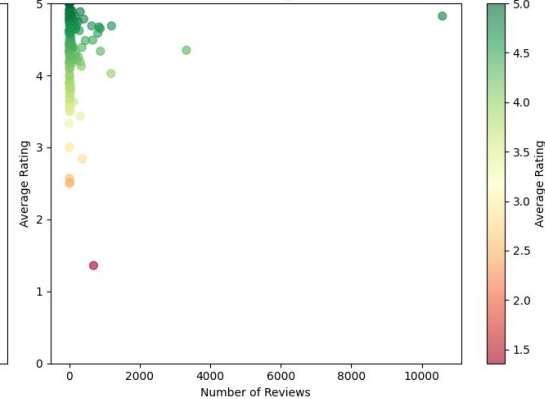
Average Rating by Top Brands (min 10 reviews)



Top 10 Product Categories



Brand Performance: Rating vs Review Count





Brand and Category Insights

- Top brands: **Clorox, Warner Home Video, Disney, L'Oréal Paris.**
- Most-reviewed categories: **Cleaning, Movies, Personal Care.**
- Popular brands also maintain strong ratings (4.5–4.8).



Modeling Approach

Steps:

1. Text preprocessing (lowercase, stopwords, lemmatization).
2. Combine ***review title + brands + categories + text*** into one feature.
3. TF-IDF vectorization.
4. Train and evaluate: Logistic Regression, Random Forest, Gradient Boosting, SVM (Support Vector Machine) .

SVM achieved best F1 = 0.93



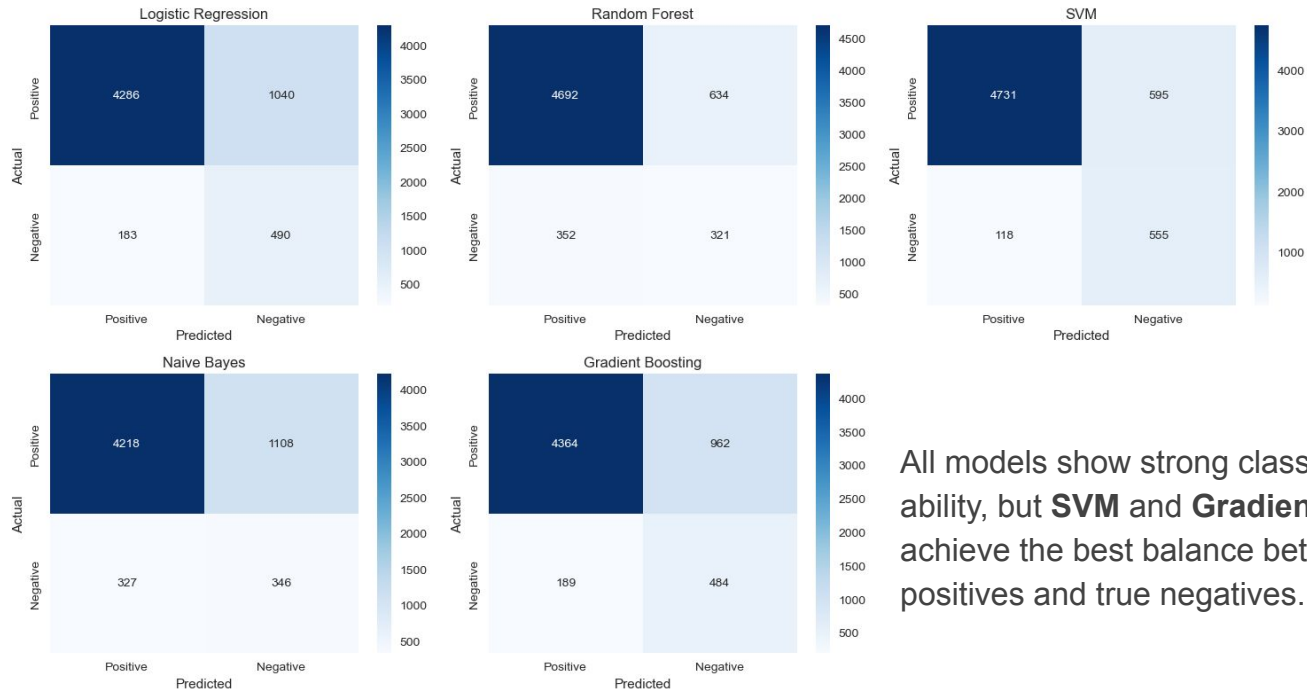
Evaluation and Metrics

	Model	Accuracy	Precision	Recall	F1-Score	Training Time (s)	Type
0	SVM	0.8808	0.9751	0.8885	0.9298	1402.80	Non-linear
1	Random Forest	0.8361	0.9306	0.8811	0.9052	2445.55	Ensemble
2	Gradient Boosting	0.8101	0.9582	0.8220	0.8849	565.82	Ensemble
3	Logistic Regression	0.7961	0.9591	0.8047	0.8751	58.78	Linear Model
4	Naive Bayes	0.7610	0.9283	0.7920	0.8547	5.76	Probabilistic



Confusion Matrix

Confusion Matrices for All Models

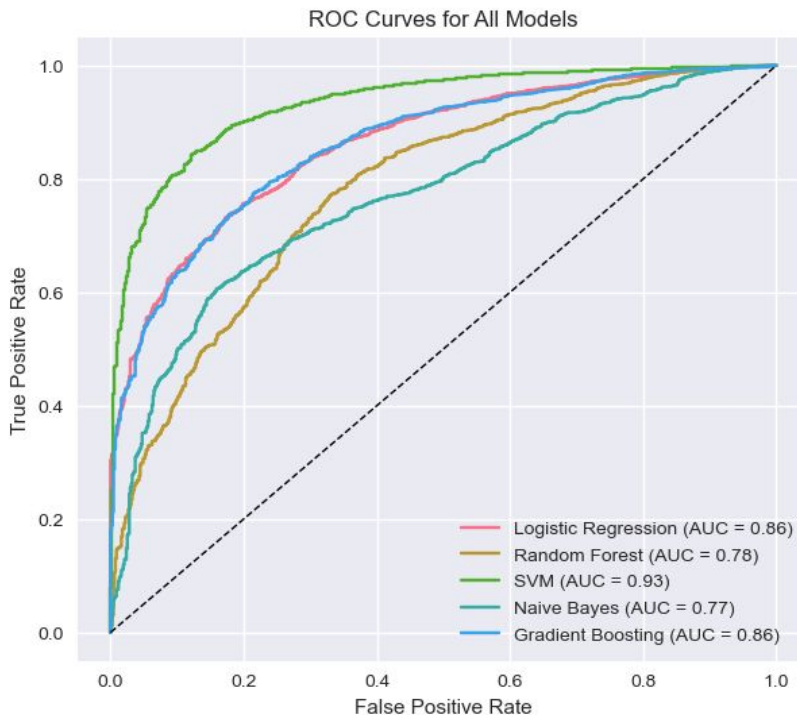


All models show strong classification ability, but **SVM** and **Gradient Boosting** achieve the best balance between true positives and true negatives.



ROC curve

Overall, the **SVM model** appears to generalize best, followed by **Logistic Regression** and **Gradient Boosting** as strong alternatives.





Recommendation Logic

Algorithm Concept:

- **For new users** → recommend top products with positive sentiment.
- **For existing users** → recommend similar brands/categories tied to previous positive feedback.



Conclusion

Outcome:

Successfully built a sentiment classification and recommendation system that combines textual emotions and structured data to improve product relevance.

Impact:

Helps businesses better understand customer satisfaction and tailor product suggestions.



Future Work

- Integrate **deep learning models** (BERT, LSTM).
- Support **multilingual sentiment detection**.
- Deploy as an **API or web app**.
- Extend to **real-time recommendation pipelines**.

QUESTION ?

NEW PRODUCT



Research

A careful study or a systematic study in order to establish facts or to discover new information.



Analysis

A process to examine something in detail in order to explain it as a basis of discussion or interpretation.



Ideas

Any thoughts, opinions, creation, suggestions or conception that is existing in the mind as to a possible cause of action.



Function

An action or activity proper to a person or thing the purpose for something which is designed for.



Ralph DJino DUMERA | | Rockson NOEL



[linkedin.com/in/ralphdumera](https://www.linkedin.com/in/ralphdumera) | | [linkedin.com/in/rockson-noel](https://www.linkedin.com/in/rockson-noel)



ralphdumera00@gmail.com | | rockson.noel@gmail.com